THE CANONICAL DELTA-FUNCTOR STRUCTURE ON THE LOCALIZATION OF COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4198-4223, following the statement at 4181-4196.

Editorial note. The two identities are substitution checks against the explicit cone formulas; recorded as routine bookkeeping.

Allowed standard prerequisites (editorial).

- Functoriality of the cone construction. lemma-functorial-cone (same chapter, cited at ~4214). Builds the canonical cone map c from (f, g).

Source reference. Original definition 0150: sources/stacks-derived/pinned-original-source.tex, source lines 516–516. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014F: sources/stacks-derived/pinned-original-source.tex, source lines 2718–2718. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. The functor $Comp(\mathcal{A}) \to D(\mathcal{A})$ defined has the natural structure of a δ-functor, with*

$$\delta_{A^\bullet \to B^\bullet \to C^\bullet} = -p \circ q^{-1}$$

with p and q as explained above. The same construction turns the functors $Comp^+(\mathcal{A}) \to D^+(\mathcal{A})$, $Comp^-(\mathcal{A}) \to D^-(\mathcal{A})$, and $Comp^b(\mathcal{A}) \to D^b(\mathcal{A})$ into δ-functors.

Proof. We have already seen that this choice leads to a distinguished triangle whenever given a short exact sequence of complexes. We have to show that given a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^\bullet & \xrightarrow{a} & B^\bullet & \xrightarrow{b} & C^\bullet & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & (A')^\bullet & \xrightarrow{a'} & (B')^\bullet & \xrightarrow{b'} & (C')^\bullet & \longrightarrow & 0 \end{array}$$

we get the desired commutative diagram of Definition 0150 (2). By Lemma 014F the pair (f, g) induces a canonical morphism $c : C(a)^\bullet \to C(a')^\bullet$. It is a simple computation to show that $q' \circ c = h \circ q$ and $f[1] \circ p = p' \circ c$. From this the result follows directly. □

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